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Comprehensive Analysis of Machine Learning Approaches for Phishing Website Detection

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ABSTRACT- **Website phishing, a notable cybercrime, involves deceiving individuals by impersonating legitimate websites. URL spoofing, a common technique in website phishing, involves creating website names. The primary objective of this paper is to accurately differentiate between legitimate and phishing sites using relevant URL attributes and different binary classification machine learning algorithms. This study compares simple, bagging, and boosting algorithms across performance analysis metrics.The best-performing models are selected for an ensemble approach. Additionally, various hyperparameter tuning techniques are assessed for the Random Forest Classifier, and Explainable AI methods like LIME and SHAP are used to interpret findings. The experiment yields optimal parameters, achieving 89.% accuracy with an ensemble model with K-Nearest Neighbors, Decision Tree, XG Boost, and Cat Boost. These findings highlight the significance of ensemble models and hyperparameter tuning in machine learning.**

Keywords-Website phishing, Classification Machine learning algorithms, Ensemble modeling, Hyperparameter tuning, Explainable AI

I. Introduction

The internet is a vast global connected network that has created a massive impact in modern society, it has not only revolutionized the fields of communication and commerce, entertainment, education, healthcare, etc. However, this establishment of interconnectivity comes with inherent threats which include privacy breaches, cyberattacks, misinformation, and digital divide issues. Website phishing is one of the most prevalent cybercrimes occurring nowadays, where the attacker crafts a website that is similar to a legitimate entity such as an e-commerce shopping platform, digital bank, or government website to deceive people into giving financial, or personal, and sensitive information, it is also be used to spread misinformation which may mislead the public and influence decision making. Attackers employ various tactics: URL spoofing being the primary method, where the attackers create URLs that remotely resemble legitimate sites, by making slight variations in the path of the website or the domain or sub-domain names. Many cyber victims struggle to differentiate between an authentic website and a fake website and end up giving sensitive information. According to a discussion at the Mobile World Congress in Barcelona, as reported by the India Times, 300 million people in India are at risk of phishing attacks, with 500,000 falling victim to these scams. [1].

From our research, it is possible to predict whether the given website is a phishing website or a non-phishing website based on its URL attributes. We performed a comparative evaluation of different classification models to enhance overall model performance and implemented Explainable AI tools like LIME, and SHAP.

The paper is structured as follows: Section 2 is overview of prior research in the field of web phishing and URL spoofing. Section 3 details the proposed workflow and various machine learning algorithm techniques utilized. Section 4 outlines the result and performance evaluation. Section 5, gives us insights about the potential future endeavours. In Section 6 we offer a potential conclusion drawn from the obtained result.

II. Related Work

Arun Kulkarni .et al[2] developed a system using 4 ML classifiers SVM, DT, Naïve Bayesian classifier, and neural network to categorize websites based on their URL. They were tested on a data set consisting of 1,353 URLsThe results indicate that the system effectively distinguished legitimate websites from phishing sites. Vahid Shahrivari .et al[3] evaluated 12 ML classifiers. RF classifier achieved the highest accuracy of 97.2%. Basnet R .et al[4]Used two publicly available datasets. A total of sixteen relevant features are used to detect phishing emails.SVM's, Neural networks SOMs, and K -Means techniques were implemented where SVM achieved the best results consistently. J. Rashid .et al [5] proposed an effective ML-based approach for phishing detection. Best performance was achieved when it was integrated with an SVM classifier, accurately distinguishing 95.66% of appropriate and phishing websites. A. Karim .et al [6] used various ML models and proposed a hybrid LSD model consisting of logistic regression, SVM, and DT. The model surpassed other models and achieved the best results. Gandotra E .et al [7] conducted a comparative analysis of ML algorithms by assessing their performance with and without feature selection. It is observed that the RF classifier produces the best results in both cases with an improvement in model building time. Jain A.K. .et al [8] Incorporate various hyperlink-specific features, which have been categorized into 12 groups to detect

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phishing sites. The efficiency is evaluated on different machine learning classification algorithms. Accuracy of 98.4% was achieved on the logistic classifier. Sahingoz O.K .et al [9] Developed a real-time phishing detection technique leveraging various ML classification algorithms and NLP-based methods . The best results were obtained by using the RF classifier, which had an accuracy of 97.98% . A.Basit .et al [10] present a novel ensemble model using the classifiers:ANN, KNN, and DT with RF Classifier. An accuracy of 97.33% was observed from this model.Benjamin McConnell .et al [11] developed a model utilizing an ensemble model, enhanced with Grid Search CV, to classify phishing websites The optimized model has an accuracy of 95.36. Rao R.S .et al [12] introduced a new classification model utilizing the features like source code, third-party services and URLs. The RF Classifier achieved the highest accuracy of 99.31%. The Principal Component Analysis Random Forest (PCA-RF) outperformed all oblique RF models, achieving an accuracy of 99.55%.Miyamoto D. et al [13] employ nine machine learning techniques including different bagging and boosting algorithms, regression trees, and neural networks. AdaBoost produced the best results and seven out of nine machine learning-based detection methods. Al-Sarem .et al [14] developed a stacking ensemble approach using a genetic algorithm (GA) to improve the parameters of various ensemble ML models. The best three performers are combined in a stacked ensemble. The experiment was conducted on three different datasets and the results showed a detection accuracy of 97.16%,98.58%, and 97.39%. Navyah Puri .et al [15] assess ML models using various elements of a URL. The SMOTE analysis technique was used to normalize the dataset. Various ML models were evaluated and accuracies were compared. It was observed that the CatBoost Classifier demonstrated the highest accuracy and F1 score. Alam, Mohammad Nazmul .et al [16] proposed a model using ML algorithms like RF Classifiers and DT. Feature selection algorithms like PCA analyze the features of the dataset extracted from the Kaggle database. Jain A.K. et al [17] introduced a machine learning-based anti-phishing system named ‘PHISH-SAFE’.. SVM portrayed accuracy greater than 90% in determining phishing websites. Almomani A.et al [18] utilized sixteen ML models to detect phishing websites, comparing the performances. These models incorporated ten semantic features crucial for detection, drawn from two datasets. According to the results, it is observed that the Gradient Boosting Classifier and RF model has an accuracy at about 97%.Ubing .et al [19] employed 3 ensemble machine learning techniques, boosting, and stacking. The classifiers are integrated to achieve the highest accuracy possible from the Decision Tree . Tyagi .et al [20] utilized a dataset from the University of California at Irvine’s Machine Learning repository, which consisted of 11,000 URLs with 6157 phishing and 4898 legitimate URL.30 attributes were utilized to forecast attacks.ML models like Decision Trees, Random Forest Classifiers, Gradient boosting, Generalized linear, and PCA were employed. Subasi A. et al [21] found their proposed classifiers highly proficient in categorizing phishing websites. They highlighted the RF Classifier as the best classifier, yielding an accuracy of 97.26%. Lochan .et al [22] employed different ML algorithms for online fraud detection. Akshitha .et al [23] used Fuzzy AHP and TOPSIS for cyber crime detection. Adam et al. [24] used ML models for fake news detection he proposed model leverages three prominent DL classifiers—RF, Naive Bayes, and Passive Aggressive classifiers—to effectively identify and mitigate fake news by analyzing patterns and inconsistencies in the content.

III. Methodology

A. Dataset Description

Phishing is an effective method for cybercriminals to deceive individuals and extract their personal and financial data. Our increasing dependence on the internet for everyday activities has created an ideal breeding ground for fraudsters to execute targeted phishing attacks.

This research aims to identify phishing websites using several ML classification models. The dataset utilized in this study was acquired from the Kaggle database.

This dataset consists of 19 features and 100077 samples. The target label comprises two classes: label '0' denotes a genuine website and label '1' denotes a phishing website.
There are 63715 samples in class "0" and 36362 in class "1". According to the given figure, Fig.1, it can be inferred that the dataset is unbalanced. To overcome this, the upscaling technique opts as shown in the figure, Fig.2.

The database acquired has to go through data preprocessing to enhance the quality and improve the results, upscaling is performed on the data to avoid data imbalance, as an uneven distribution of classes and categories may lead to wrong predictions, this ensures that the algorithm runs smoothly once data has been pre-processed. Subsequently, various supervised learning classifier algorithms have been considered. The methods include

- Single classifier
- Bagging and Boosting
- Ensemble

These models help in prediction by categorizing and grouping the data into distinct classes or categories.

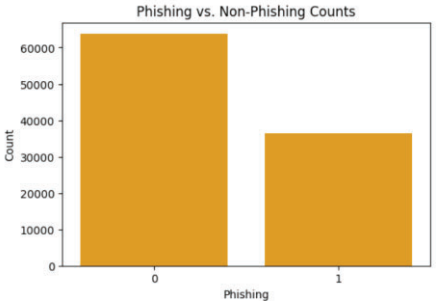


Fig. 1. Dataset before upscaling

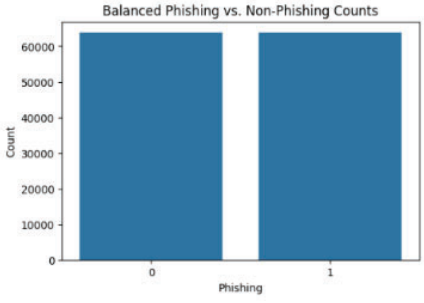


Fig. 2. Dataset after upscaling

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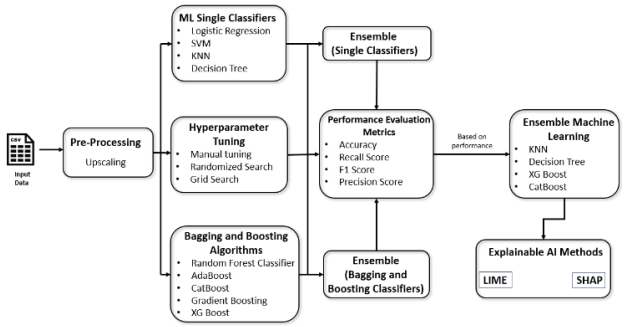


Fig. 3. Proposed Workflow

B. Proposed Workflow

A single classifier model encompasses 4 algorithms that are Logistic Regression, KNN, SVM, and DT. Logistic regression output is predicted based on the weighted sum of the features, it is a simple and efficient model especially when dealing with problems with two-class classification.In KNN, the prediction of new data points is done by considering the nearest data points or neighbors in the feature space. The distance between the data points is calculated to find similarities.Here the ”K” in KNN refers to the neighbors which have to be considered to make a prediction. SVM algorithm is well suited to classify the data points into 2 separate groups, this is done by seeking an optimal hyperplane in a 19-dimensional hyperplane that divides the data into 2 classes in the feature space, when the new data is introduced SVM determines which side of hyperplane it falls on in the feature space based on, the positioning the data point is classified in one of the two classes. In a DT, a tree-like structure is created using multiple subsets of input features. The splitting of the data is determined by specific criteria, such as the Gini impurity and entropy methods. These methods guide the process of partitioning the data, ensuring that each split optimally separates the dataset into more homogeneous subsets. The evaluation of this method is done based on accuracy, precision, F1 score, and recall.

The second model to be considered is bagging and boosting, It help in improving the accuracy by reducing the overfitting and variance. These methods involve training multiple models and combining their predictions for a robust final output. Algorithms that come under Bagging and Boosting are RF, ADA Boost, XG Boost, Gradient Boosting, and CAT Boosting. Under bagging one prominent algorithm is RF .It constructs various trees using the training data. During prediction, it aggregates the prediction from individual trees and outputs the mode of all individual tree results. ADA boost employs iteratively training the weak classifiers, and their outputs are combined to enhance the performance. The gradient boosting ensembles various weak models sequentially to build a better model. XG boost is an optimized implementation of gradient boosting, it uses advanced techniques to improve speed and performance.CAT boost is generally used for datasets having mixed categorical values, it is used to extend the capabilities of gradient boosting.

Ensemble methods are techniques used in machine learning that integrate various individual models for effective performance. Ensemble the results of all the algorithms that come under the single classifier and bagging and boosting are performed to improve the performance that could not be achieved by the individual models. A hyperparameter is applied to the RF algorithm to improve the model. The optimization of the model can be achieved by fine-tuning the key parameters associated with the Random Forest classifier.

Explainable AI is used to give reasoning behind the decision-making made by an AI. LIMEand SHAP are one of the techniques that provide insights into how individual features influence the decision-making process.

IV. Results and Discussions

A. Single classifier results

The initial phase of our observation was conducted using various single MLclassifiers to detect website phishing. The results are tabulated in the following table, Table I.

TABLE I.

Algorithm	Accuracy	F1	Recall	Precision
Logistic	0.844	0.7795	0.8329	0.8941
SVM	0.8851	0.8773	0.8839	0.8906
DT	0.8942	0.894	0.894	0.8939
KNN	0.8472	0.7876	0.8372	0.8934



Fig. 4. Performance graph of Single Classifiers

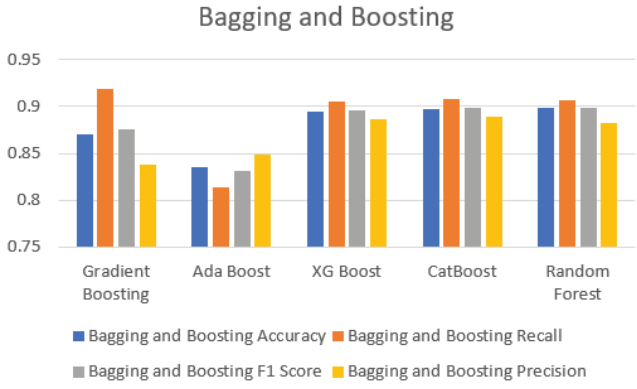


Fig. 5. Performance graph of Bagging and boosting classifiers

Among the classifiers assessed, the DT classifier, with an accuracy of 89.4%, outperformed other models It is capable of handling non-linear correlations between variables, and imbalanced data, making it particularly suitable for the dataset. Furthermore, its capacity to be able to handle large datasets and perform classification with minimal computation or complexity contributes to its competence.

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B. *Bagging and Boosting Classifiers*

To improve the observation and findings, multiple, bagging and boosting models were introduced.

Comparing the graphs Fig. 4 and Fig. 5, it is observed that these classifiers have outperformed the single classifiers. Bagging and boosting models, specifically XG Boost, CatBoost, and Random Forest Classifier have demonstrated promising accuracies of 89.4%,89.7%, and 89.8% respectively. Their ability to reduce variance, improve generalization, and enhance accuracy, especially in complex datasets significantly contributes to improved results.

C. *Proposed Ensemble Model*

Ensemble models combine predictions of several base models that capture diversity, improving overall performance. Building upon the success of individual models, a proposed ensemble model was introduced to enhance performance further. Three ensemble models were constructed, each incorporating different classifiers: a model solely comprising of single classifiers, another comprised solely of bagging and boosting techniques, and a third model integrating the best-performing models from the preceding categories. Remarkably, the ensemble model combining both single classifiers and bagging methods achieved the highest accuracy of 89.9%.

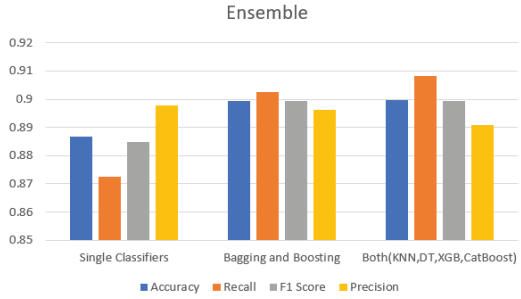


Fig. 6. Performance graph of Proposed Ensemble model

D. *Hyperparameter Tuning*

Expanding upon the encouraging outcomes achieved with the Random Forest Classifier, the accuracy was systematically refined using different hyperparameter tuning techniques. Initially, manual hyperparameter tuning resulted in an accuracy of 89.29%. Following iterations using Randomized Search CV and Grid Search CV boosted the accuracies to 89.47% and 89.39% respectively. The figure, Fig. 7 illustrates the improvement of model accuracies across various tuning techniques. The optimal configuration that maximized the accuracy is as follows: ‘Criterion' as 'gini', a ‘max_depth’ of 890, 'max_features' as 'sqrt', 'min_samples_split' as a set of [3, 5, 7], 'min_samples_leaf' as a set [0, 1, 2, 3, 4], 'n_estimators' as a set of [513, 713, 813].

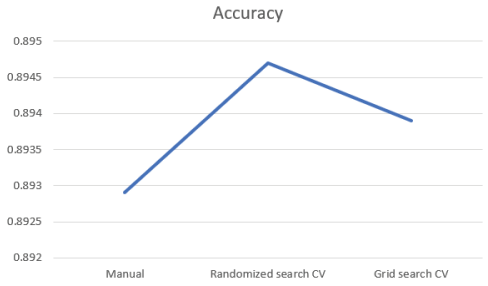


Fig. 7. Accuracy of different Hyperparameter Tuning methods

E. *Interpretation of results using Explainable AI Methods*

The study utilized Explainable AI methods like LIME and SHAP explainer models to interpret and offer insights into the findings derived from the machine learning classifier models.

1) *LIME Explainer*

Local Interpretable Model-Agnostic Explanation (LIME) is a technique used for interpreting the results from the machine learning model. Considering any one of the instances from the entire dataset, LIME shows how the features are contributing to decision-making.

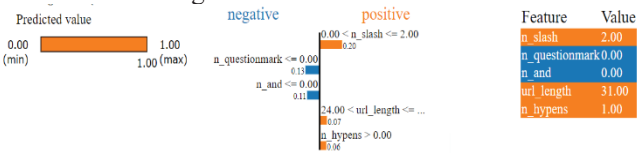


Fig. 8. Interpretation of outcomes using LIME for sample 31891

Considering the local instance of 31891 from the dataset, figure Fig. 8 shows that the features ‘n_hypens’,’url_length’, and ‘n_slash’ are influencing the prediction positively whereas the attributes, ‘n_questionmark’ and ‘n_and’ are influencing the prediction negatively. With a significant value of 31.00, the feature ‘url_length’ has a major impact on the final prediction. All the features collectively predict the given instance as the class ‘1’, indicating that the website is not legitimate.

Similarly, for the 85th instance, the features ‘url_length’ and ‘n_hypens’ contribute positively and the features ‘n_slash’,’n_equal’, and ‘n_and’ contribute negatively towards the prediction as shown in the figure, Fig.9.. Therefore, the instance is predicted as the class ‘0’, indicating that the website is legitimate.

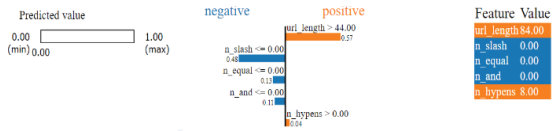


Fig. 9. Interpretation of outcome using LIME for instance 85

2) *SHAP Explainer*

SHAP values play an essential role in understanding the importance of every individual feature when predicting an instance. By distributing the prediction value among the input features in an efficient manner, these values provide essential insights into the relative significance of every feature in the process.

A study of the graph, Fig. 10., shows the features and their importance, where we can conclude that ‘url_length’ is one of the most significant features for decision-making. The figure represents all the 19 features of the dataset based on their weights assigned by the models. The features of test samples are showcased as SHAP is a global interpreter.

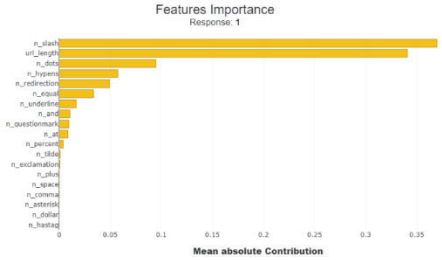


Fig. 10. Interpretation of results using SHAP for feature importance(global)

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A subset of the dataset consisting of 50 samples is created. It is analyzed from Fig. 11 that the features deemed highly important in the test samples remain consistent with minute variations.

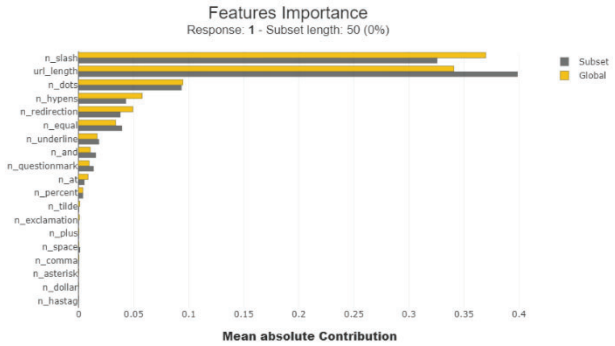


Fig. 11. Interpretation of results using SHAP for subset

Figure Fig. 12 displays the contribution of the 'url_length' feature to individual test samples. All the data points above the horizontal line correspond to the samples where the impact of this feature is positive, predicting as a non-legitimate website, whereas all the points below the horizontal line represent the samples for which the feature is negative, predicting as a legitimate website. The contribution of the 'url_length' feature toward ID 3451 from the testing dataset is 0.4141.

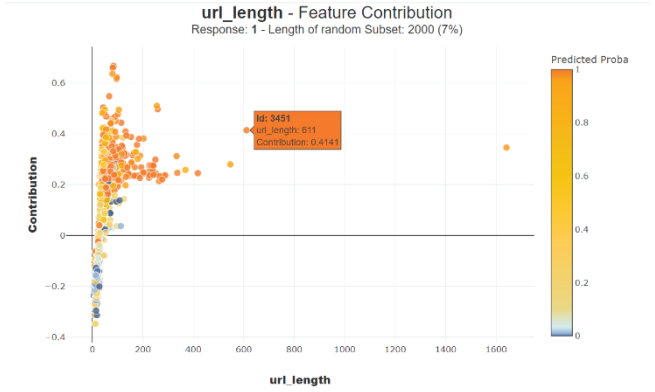


Fig. 12. Interpretation of results using SHAP for a individual attribute

The local interpretation of the sample with ID 5276 can be visualized in the figure Fig. 13. As the number of features contributing to the positive output ,represented by yellow bars exceeds those contributing to the negative output, represented by grey bars, the website is classified as phishing.

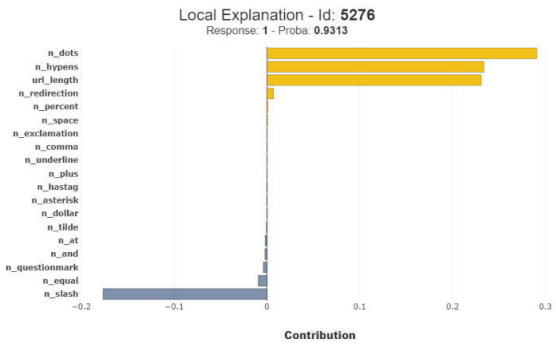


Fig. 13. Interpretation of results using SHAP for a single sample

V. CONCLUSION AND FUTURE WORK

Nine classifiers were implemented on the phishing website dataset. To increase the accuracy and efficiency of the detection, ensemble models and hyperparameter tuning techniques were introduced. The findings indicate good ensembling performance for classifiers KNN, DT, XGBoost, and CatBoost. There is no guarantee that the combination of various classifiers will always perform better than an individual classifier. Additionally, a Randomized Search CV for the Random Forest Classifier produces favorable results.

Although the obtained results are satisfactory, efficiency can be increased using advanced technologies such as deep learning and parallel processing for a huge dataset. The performance of this model can also be improved by inculcating other phishing detection techniques along with machine learning. Moreover, new subsystems can be performed for shorter URLs.

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